

Gaurav Lochab

✉ gauravlochab487@gmail.com | 📞 +91-8222022505
🐙 github.com/gauravlochab | 🔗 linkedin.com/in/gauravlochab487

SKILLS

Languages: Python (Primary), SQL, Bash, JavaScript

Infrastructure & MLOps: Docker, Kubernetes, AWS (Lambda, SageMaker), Azure, Apache Airflow, vLLM, FastAPI, Containerized Inference, Distributed Systems, CI/CD, ETL Pipelines, Linux

ML & AI: LLMs, RAG Systems, LLM Fine-Tuning (QLoRA, ORPO), LLM Evaluation & Benchmarking, Semantic Search, Elasticsearch, Qdrant, Embeddings, Model Quantization (INT8), Generative AI, LangChain, LangGraph, Hugging Face

WORK EXPERIENCE

Apolis, Remote

October 2025 – Present

Senior AI / Machine Learning Engineer

- Architected an end-to-end **ML retrieval and ranking platform** in **Python** across **2.7M+ candidate resumes**, using **Elasticsearch BM25**, dense embeddings, and **distributed scoring pipelines**.
- Built **containerized embedding-driven semantic search APIs** using **vLLM inference** and **FastAPI**, deployed on **Docker** with semantic scoring, synonym expansion, and guided JSON generation.
- Designed **MLOps pipelines** for resume and job description extraction at scale, including **evaluation frameworks** benchmarking field accuracy, reasoning consistency, and hallucination detection across model versions.
- Designed a **two-stage retrieval and ranking system** combining high-recall search with ML scoring across skill overlap, role similarity, and experience alignment, significantly improving relevance.
- Fine-tuned **Qwen2.5 Instruct** models using **QLoRA (4-bit NF4 quantization)** and implemented **teacher-student distillation pipelines** to improve extraction accuracy while reducing inference cost and latency.

Valory

June 2024 – October 2025

Senior AI Developer

- Built **distributed autonomous agent systems** in **Python** for portfolio management, scaling to **5M+ requests** and **\$10M+ managed assets** with **<200ms latency**.
- Engineered **LangGraph-based ML pipelines** with state-machine orchestration and **Apache Airflow** for production scheduling, integrating **GPT-4**, **Claude**, and fine-tuned **Llama models**.
- Operated production AI infrastructure with **CI/CD pipelines**, **containerized deployment**, and observability across multi-protocol blockchain integrations.

Smarter, London, UK (remote)

Oct 2022 - May 2024

Senior Machine Learning Engineer

- Engineered **ML serving pipelines** with **4x model size reduction** and **2x latency improvement** via **INT8 quantization**, deploying **containerized inference** using **Docker** and **vLLM** on Linux while maintaining **95% accuracy**.
- Developed an **LLM evaluation pipeline** using **Llama3-8B** fine-tuned with instruction tuning and **ORPO preference alignment**, benchmarking and replacing underperforming models to improve accuracy by **15%**.
- Built production **RAG agent** using **Llama3-8B**, **FastAPI**, **LangChain**, and **Qdrant**; implemented **query expansion and re-ranking** achieving **30% improved retrieval relevance**.

Pibit.ai, Gurugram, India

June 2020 - Sept 2022

Founding Team / Machine Learning

- Built **Python-based ETL pipelines** for large-scale document processing on **AWS**, improving throughput by **30%** while reducing processing errors by **25%**.
- Developed **document intelligence pipelines** using NLP and computer vision, improving structured data extraction accuracy by **35%**, deployed via **Docker**, **AWS Lambda**, and **SageMaker**.
- Designed and deployed **fraud detection systems** combining supervised anomaly detection, unsupervised clustering, and rule-based pattern analysis, with **containerized APIs** on AWS.

EDUCATION & RESEARCH

USICT Delhi

Aug 2016 - May 2020

B.E. in Computer Science and Engineering

CGPA: 8.1/10

- **Research Paper Published** — International Advanced Computing Conference (IACC) 2022
- **Mentor, Scaler Academy** — Guided students in machine learning, data engineering, and applied ML systems